

A candidate proof of Conjecture 9.5 in Random Embeddings of Graphs: The Expected Number of Faces in Most Graph...

github.com/graph-theory-AI — machine-generated candidate writeup

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Abstract

A fixed-vertex decomposition proves $E[F_n] \leq \ln n + O(1)$, with the stronger upper bound $\ln n + 1/4 + \gamma/2 + o(1)$. This candidate proof was generated by GPT-6 Astra in a single model pass for the Graph Theory AI project. It was selected from the campaign because the model marked it `would_publish: true`; it has not been checked by the adversarial referee, a human mathematician, or a novelty search.

1 The problem and the claimed result

This document concerns [Conjecture 9.5](#) from *Random Embeddings of Graphs: The Expected Number of Faces in Most Graph...*

1.1 Problem statement

Conjecture. The expected number of faces in a non-orientable random embedding of the complete graph K_n is at most $\ln(n) + O(1)$.

1.2 Catalog context

The authors extend their investigation to non-orientable surfaces, where a random embedding is defined by choosing a random rotation system together with a random signature assigning each edge a sign independently with probability $1/2$. Computer simulations support a logarithmic behaviour analogous to the orientable case.

1.3 Claimed result

Model verdict: proved (confidence: high).

A fixed-vertex decomposition proves $E[F_n] \leq \ln n + O(1)$, with the stronger upper bound $\ln n + 1/4 + \gamma/2 + o(1)$.

The model's own caveat: The proof uses exactly the stated uniform-rotation, independent-fair-signature model; literature priority has not been checked.

2 Model writeup

The text below is the model output verbatim, apart from moving its machine-readable verdict block into the summary above and shifting Markdown heading levels for this document. Mechanical TeX defects and equation tags were normalized where needed for compilation.

2.1 A fixed-vertex proof

Let F_n denote the number of faces in the random signed-rotation embedding of K_n specified in the question.

The key is to separate faces meeting one fixed vertex from faces avoiding it. Each class has expected size at most $\frac{1}{2} \ln n + O(1)$. For the avoiding faces, an unvisited vertex supplies an escape option at every exploration step. This controls local congestion and removes the long-face obstruction in the previous attempt.

The flag-matching representation from that attempt is valid and is justified below. Its short-face enumeration is not needed.

2.1.1 Theorem

As $n \rightarrow \infty$,

$$\mathbb{E}F_n \leq \ln n + \frac{1}{4} + \frac{\gamma}{2} + O\left(\frac{(\ln n)^2}{n}\right),$$

where γ is Euler's constant. In particular,

$$\mathbb{E}F_n \leq \ln n + O(1).$$

All logarithms below are natural unless a base is indicated.

2.1.2 1. The matching model and its exposure rule

Put $d = n - 1$. For each incidence of an edge uv at u , introduce two flags

$$(u, v, 0), \quad (u, v, 1).$$

Thus there are $2d$ flags at each vertex and $2n(n - 1)$ flags altogether.

Use the following matchings:

- J_u pairs the two flags of each incidence at u .
- A pairs corresponding flags at opposite ends of an edge:

$$(u, v, s) \longleftrightarrow (v, u, s).$$

- V_u is a matching on the flags at u , describing the arcs between consecutive edge attachments around its vertex-disc.

The boundary components of the resulting ribbon graph are exactly the connected components of

$$A \cup V, \quad V = \bigcup_u V_u.$$

A face of length k contains k matching edges from A , k from V , and $2k$ flags. Consequently,

$$\sum_{\text{faces } f} |f| = n(n - 1).$$

2.1.2.1 Distribution of the local matchings

Let

$$\mathcal{H}_d = \{W : W \text{ is a perfect matching and } J_u \cup W \text{ is one alternating } 2d\text{-cycle}\}.$$

In the stated random-embedding model, we may take the V_u to be independent and uniform on $\mathcal{H}_d$, with A fixed as above.

Here is a verification. Choose independent fair bits $b_{u,e}$ at all incidences, and define the signature of $e = uv$ by

$$\varepsilon_e = b_{u,e} \oplus b_{v,e},$$

with an immaterial fixed adjustment if the opposite sign convention is used. These signatures are independent and fair. Relabelling the flags at incidence (u, e) using $b_{u,e}$ makes the edge matching canonical.

At a vertex, a cyclic order and the d incidence bits specify an element of $\mathcal{H}_d$. Every element has exactly two such representations, corresponding to the two directions around its alternating cycle. Hence

$$|\mathcal{H}_d| = 2^{d-1}(d-1)!,$$

and the local matchings are independent and uniform, as asserted.

2.1.2.2 Local exposure lemma Suppose $r < d$ pairs of V_u have been exposed, forming a matching Q_u .

Because $J_u \cup V_u$ is one cycle, $J_u \cup Q_u$ consists of $d - r$ alternating paths. Its endpoints are exactly the flags whose V_u -mates are still unexposed.

If $r \leq d - 2$, then, for any such flag x , its mate is uniform among

$$2(d - r - 1) \quad (1)$$

possible flags: all the other free flags except the opposite endpoint of the same $J_u \cup Q_u$ -path. If $r = d - 1$, the last pair is forced.

Indeed, contracting the alternating paths reduces the conditional completion problem to a uniform element of $\mathcal{H}_{d-r}$. Joining two different paths is allowed, while closing a proper local cycle is forbidden. All allowed partners have the same number of completions.

This rule remains valid under adaptive exposure using only previously revealed information.

2.1.3 2. Faces meeting a fixed vertex

Fix a vertex o . Write

$$F_n = F_n^{\text{hit}} + F_n^{\text{avoid}},$$

where the two terms count faces meeting and avoiding o , respectively.

Expose all local matchings except V_o . The components of the exposed boundary graph are:

- cycles, which are precisely the faces avoiding o ;
- paths whose endpoints are flags at o .

The latter paths induce a perfect matching P on the $2d$ flags at o . Conditional on this exposure, V_o is still uniform on $\mathcal{H}_d$, and the faces meeting o correspond exactly to the components of $P \cup V_o$.

Expose V_o pair by pair. After $r \leq d - 2$ pairs have been exposed, select any free flag x . At most one possible mate closes a component of P together with the exposed pairs: the opposite endpoint of the current P -path containing x . By (1), the conditional probability of completing a component is at most

$$\frac{1}{2(d - r - 1)}.$$

The final forced pair completes at most one component. Every component is counted when its last pair is exposed. Therefore

$$\mathbb{E} F_n^{\text{hit}} \leq 1 + \frac{1}{2} H_{d-1} = 1 + \frac{1}{2} H_{n-2}. \quad (2)$$

The remainder of the proof bounds F_n^{avoid} .

2.1.4 3. Exploring one face while avoiding o

We now use a different exposure order, analyzing one starting flag at a time and revealing no other faces.

Choose a flag a_0 on an edge not incident with o , and put

$$b_0 = A(a_0).$$

Start with the A -edge $a_0 b_0$, treating a_0 as the active flag and b_0 as a fixed terminal.

At each step:

1. reveal the V -mate z of the active flag;
2. if $z = b_0$, the face closes;
3. otherwise, make $A(z)$ the new active flag;
4. stop if the new active flag is at o .

Until closure, all exposed V -pairs lie on one path with endpoints the active flag and b_0 . All other components of A together with these exposed pairs are isolated A -edges. Thus every query is to a previously unexposed mate.

If the trace closes after k queries, its face has length k .

2.1.4.1 The escape probability Suppose the process is alive, its active flag is at $u \neq o$, and r_u pairs have been exposed at u .

Neither flag of incidence (u, uo) has been used. Otherwise the trace would already have visited o . Consequently those two flags form an isolated J_u -path, different from the path containing the active flag. Both are allowed choices in the local exposure lemma.

In particular, there are at least two unfinished local paths, so the forced-last-pair case cannot occur while the trace avoids o . The conditional probability of moving next to o is exactly

$$p_o = \frac{2}{2(d - r_u - 1)} = \frac{1}{n - 2 - r_u}. \quad (3)$$

For any other vertex w , there are at most two flags at u whose A -partners lie at w . Some might already be unavailable, and choosing the terminal may instead close the face. Thus the conditional probability p_w of making w the next active vertex satisfies

$$p_w \leq p_o. \quad (4)$$

Let T be the number of queries until closure or the first visit to o . Set

$$q = 1 - \frac{1}{n-2}.$$

Equation (3) gives

$$\boxed{\Pr(T > t) \leq q^t.} \quad (5)$$

2.1.4.2 A geometric bound on repeated visits Fix $w \neq o$. While the trace is alive, let $N_w(t)$ count its active visits to w , including the current active position and the initial position if applicable.

Define

$$M_t = \begin{cases} 2^{N_w(t)}, & t < T, \\ 0, & t \geq T. \end{cases}$$

Let p_{cl} denote the conditional probability of closing at the next query. While alive,

$$\mathbb{E}[M_{t+1} \mid \mathcal{F}_t] = M_t(1 - p_o - p_{\text{cl}} + p_w) \leq M_t$$

by (4). Hence M_t is a nonnegative supermartingale, with $M_0 \leq 2$.

Stop it on reaching L active visits to w , or on termination. There are at most $n(n-1)$ queries, so bounded stopping gives

$$\boxed{\Pr(N_w(t) \geq L \text{ for some } t < T) \leq 2^{1-L}.} \quad (6)$$

This controls local congestion throughout the entire trace, not merely up to a predetermined length.

2.1.5 4. Bounding all faces avoiding o

For sufficiently large n , choose

$$R = \lceil 4 \log_2 n \rceil, \quad D = n - 2 - R > 0. \quad (7)$$

Call an avoiding face **good** if it uses at most R local pairs at every vertex, and **bad** otherwise.

Let S be the flags on edges not incident with o . Then

$$|S| = 4 \binom{n-1}{2} = 2(n-1)(n-2). \quad (8)$$

2.1.5.1 4.1. Bad avoiding faces contribute $o(1)$ If the face of a starting flag avoids o and is bad, its trace reaches $R+1$ active visits to some vertex before terminating. By (6) and a union bound,

$$\Pr(\text{the starting flag lies in a bad avoiding face}) \leq (n-1)2^{-R}. \quad (9)$$

For a flag a , let $L(a)$ be its face length. Every length- k face has $2k$ flags, so the number B of bad avoiding faces is exactly

$$B = \sum_{a \in S} \frac{\mathbf{1}_{\{a \text{ lies in a bad avoiding face}\}}}{2L(a)}.$$

Using $L(a) \geq 1$, (8), and (9),

$$\mathbb{E}B \leq (n-1)^2(n-2)2^{-R} \leq n^3 2^{-R} \leq \frac{1}{n}. \quad (10)$$

No independence between traces from different starting flags is used.

2.1.5.2 4.2. Closing probabilities for good faces Fix a starting flag, and let w_0 be the vertex containing its terminal b_0 .

For the trace to close at query k , it must:

- remain alive after query $k - 2$;
- at query $k - 1$, move to w_0 ;
- at query k , choose the particular terminal flag b_0 .

When a revealed history has used at most R local pairs at every vertex, the exposure rule gives at least $2D$ possible partners at every relevant query. Therefore:

- the probability of moving next to w_0 is at most $2/(2D) = 1/D$;
- once there, the probability of choosing b_0 is at most $1/(2D)$.

These are bounds conditional on the revealed history, **not** on eventual goodness. To estimate a good-face event, discard continuations whose histories cease to satisfy the load restriction. Applying the two bounds successively, and then (5), gives, for $k \geq 2$,

$$\Pr(\text{the starting flag lies in a good avoiding face of length } k) \leq \frac{q^{k-2}}{2D^2}. \quad (11)$$

This estimate applies to every possible face length.

2.1.5.3 4.3. Summation over lengths For $n \geq 3$, there are no faces of length one or two.

Length one is impossible because K_n has no loops. A length-two face would have to use both sides of one edge uv . At each endpoint, this would require V_u or V_v to pair the two flags of that same incidence. Such a pair would form a proper alternating cycle with J_u or J_v , contrary to membership in $\mathcal{H}_d$ when $d > 1$.

Let G_k count good avoiding faces of length k . Rooting at flags, (8) and (11) give

$$\begin{aligned} \mathbb{E}G_k &= \frac{1}{2k} \sum_{a \in S} \Pr(a \text{ lies in such a face}) \\ &\leq \frac{(n-1)(n-2)}{2D^2} \frac{q^{k-2}}{k}. \end{aligned}$$

Together with (10),

$$\mathbb{E}F_n^{\text{avoid}} \leq \frac{(n-1)(n-2)}{2(n-2-R)^2} \sum_{k=3}^{\infty} \frac{q^{k-2}}{k} + \frac{1}{n}. \quad (12)$$

The series is explicit:

$$\begin{aligned} \sum_{k=3}^{\infty} \frac{q^{k-2}}{k} &= q^{-2} \left(-\ln(1-q) - q - \frac{q^2}{2} \right) \\ &= q^{-2} \left(\ln(n-2) - q - \frac{q^2}{2} \right) \\ &= \ln n - \frac{3}{2} + O\left(\frac{\ln n}{n}\right). \end{aligned} \quad (13)$$

Also, because $R = O(\ln n)$,

$$\frac{(n-1)(n-2)}{2(n-2-R)^2} = \frac{1}{2} + O\left(\frac{\ln n}{n}\right). \quad (14)$$

Consequently,

$$\mathbb{E}F_n^{\text{avoid}} \leq \frac{1}{2} \ln n - \frac{3}{4} + O\left(\frac{(\ln n)^2}{n}\right). \quad (15)$$

2.1.6 5. Conclusion

From (2),

$$\mathbb{E}F_n^{\text{hit}} \leq 1 + \frac{1}{2}H_{n-2} = \frac{1}{2} \ln n + 1 + \frac{\gamma}{2} + O(1/n).$$

Adding (15) proves

$$\mathbb{E}F_n \leq \ln n + \frac{1}{4} + \frac{\gamma}{2} + O\left(\frac{(\ln n)^2}{n}\right).$$

This establishes the conjecture. The argument was stated for sufficiently large n ; finitely many smaller values are absorbed into the $O(1)$ in the conjecture.

2.1.6.1 If one conditions on non-orientability The stipulated independent-signature model also produces orientable embeddings with a very small probability. If “non-orientable random embedding” is instead intended to mean conditioning on non-orientability, the conclusion remains unchanged.

For a connected graph, precisely 2^{n-1} signatures are orientable: these are obtained by switching vertex orientations from an all-orientable signature. Thus, for K_n , the orientability probability is

$$p_n = 2^{n-1-\binom{n}{2}} = 2^{-(n-1)(n-2)/2}.$$

Since $F_n \geq 0$,

$$\mathbb{E}[F_n \mid \text{non-orientable}] \leq \frac{\mathbb{E}F_n}{1 - p_n},$$

which preserves the displayed asymptotic upper bound.

The long-face gap is therefore closed: faces meeting the fixed vertex are controlled by a single local matching exposure, while avoiding faces admit both an exponential survival bound and a geometric local-congestion bound. No additional assumption about mixing or long facial walks is required.

A Provenance and review status

UNREFEREED MODEL OUTPUT. This document was selected solely because GPT-6 Astra labelled its own result `would_publish: true` and returned `proved` or `disproved`. It has not passed the adversarial LLM referee used for the earlier Sol campaign, has not been checked by a human mathematician, and has not been checked for novelty. Treat every mathematical and bibliographic claim as unverified.

Catalog id	2211.01032__03
Catalog entry	Conjecture 9.5
Source corpus	arxiv
Campaign leg	selected second attempts (attacks_retry)
Source paper / entry	Random Embeddings of Graphs: The Expected Number of Faces in Most Graph...
Model verdict	proved (confidence: high)
Selection criterion	<code>would_publish: true</code> and verdict <code>proved</code> or <code>disproved</code>

Model	gpt-6-astra, reasoning effort max, mode=pro, flex service tier
Original artifact	attacks_retry/2211.01032__03/output.md
Independent review	None
arXiv	arXiv:2211.01032
Previous attempt	attacks/2211.01032__03: gpt-5.6-sol verdict partial
Model's caveats	The proof uses exactly the stated uniform-rotation, independent-fair-signature model; literature priority has not been checked.

The generated Markdown and LaTeX sources for this PDF are stored under to_review_astra/src/2211.01032__03/.